- Egg white contains protein. When whipped it forms films and entraps air. On heating it coagulates to produce rigidity.

6. Shortenings:

- Shortenings are fats and oils.
- Butter, margarine and hydrogenated oils are the most common shortenings used in baking.
- It acts as tenderizers.
- It melts and releases air bubbles which will help in the leavening action of baking powder and expanding steam.

DO YOU KNOW?

Transfat -
A fat resulting from the partial hydrogenation process that converts the cis form of unsaturated fats into trans form of unsaturated fats.
Transfats are found in Margarine and Baked goods.

7. Sugar:

- Sugar is a tenderizer in baked foods.
- It is necessary for yeast growth and indirectly aids the fermentation process.

DO YOU KNOW?

Caramelisation -
It is a non-enzymatic browning process resulting due to the heating of foods containing sugar into a nutty flavor and brown colour
Sugar caramelizes at 180°C–210°C

- Brown colour of the crust is due to the Maillard reaction between the protein and sugar which occurs during baking.
- Sugar influences the tenderness and the volume of baked products. Honey and glucose are also used in baked products.

8. Other Ingredients:

Milk powder and skimmed milk are used in bread and bun making. It increases the nutritive value of bread. It improves flavour and gives a brown crust.

Oxidising agents like potassium bromate, potassium iodate and calcium peroxide are used to improve the handling characteristics of the dough and specific volume and texture of the finished products.

Salt has a retarding effect on yeast fermentation. Salt is used as a taste enhancer and as a preservative.

Water is important for gluten formation. It dissolves sugar and salt and serves as a dispersion media for yeast cell.

5.2.1 Principles of Baking

Baking is a heating process in which many reactions occur at different rates. Some of these reactions include the following:

1. Evolution and expansion of gases.
2. Coagulation of gluten and eggs.
3. Gelatinization of starch.
4. Partial dehydration from evaporation of water.
5. Development of flavours.

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